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Acoustic ringing effects in pulsed nuclear magnetic resonance probes

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The troublesome spurious ringing phenomenon found in pulsed nuclear magnetic resonance probes is explained in terms of the electromagnetic generation and detection of ultrasonic waves. A few techniques for eliminating this problem are discussed.

INTRODUCTION

In pulsed nuclear magnetic resonance (NMR) studies of solids, large line widths and correspondingly short free induction decay (FID) signals are generally encountered. In order to observe these signals the system must recover quickly after the pulse. A phenomenon frequently found to obscure the FID signal is a high- Q spurious ringing signal following the rf pulse. This ringing may persist for milliseconds and its amplitude depends on the static magnetic field H_0 , the rf field H_1 , and the length of the rf pulse, while its shape depends only on the master oscillator frequency. Clark¹ attributed this "coil disease" to acoustic responses within the coil or sample and mentioned a few measures which could be taken to reduce the problem. Speight, Jeffrey, and Courtney² later determined that while this effect may occur, the major source of the problem was the presence of solid metal surfaces near the coil, such as the probe walls or pole faces of the magnet. They suggested that eddy currents induced at these surfaces by the rf pulse somehow interact with the static magnetic field in such a way as to feed energy back into the coil. A simple shield was then described which in most cases effectively removed the spurious signal.

Our investigations have shown that the acoustic ringing in the coil and the eddy current effect in the probe wall are closely related in that they are both manifestations of what is known as the direct electromagnetic generation of acoustic waves.³ This process is well understood and is in fact used as a source of ultrasonic waves in cases where direct contact of a transducer with the sample is undesirable.

I. THEORY

The electromagnetic generation of ultrasonic waves was first observed at helium temperature in single crystal disks of Bi by Gantmakher and Dolgoplov⁴ and Al by Larsen and Saermark.⁵ Betjemann *et al.*⁶ observed the same phenomenon over a wide temperature range in various metals, including rods of commercial Cu and Al at room temperature. The classical explanation is given elsewhere,^{7,8} so only a brief sketch shall be presented here.

An rf magnetic field of angular frequency ω which is linearly polarized in the xy plane will induce an eddy

current $\mathbf{j}(z) \exp(i\omega t)$ within the skin depth δ of the metal, as shown in Fig. 1. If there is a static magnetic field $\mathbf{H}_0$ along the z direction, the electrons will experience a Lorentz force given by

$$\mathbf{F}(z) = \mathbf{j}(z) \times \mathbf{H}_0 \exp(i\omega t)/nc, \quad (1)$$

where n is the number of electrons per unit volume. In order for the interior of the metal to remain shielded from the applied rf field, this force must also be experienced by the lattice.⁸ Therefore, $\mathbf{F}(z)$ is the driving force for a shear wave which propagates through the metal in the z direction. [If $\mathbf{H}_0$ were parallel to the rf field $\mathbf{H}(z)$, $\mathbf{F}$ would be along the z direction and longitudinal waves would be generated.] The equation of motion of the ions (neglecting acoustic attenuation within the metal) is

$$\frac{\partial^2 \mathbf{u}}{\partial t^2} - s^2 \frac{\partial^2 \mathbf{u}}{\partial z^2} = |\mathbf{j}(z) \times \mathbf{H}_0| \exp(i\omega t)/dc, \quad (2)$$

where the ionic displacement $\mathbf{u}$ is parallel to $\mathbf{F}$, and $d = nM/Z$ is the density of the metal, Ze the charge per ion, M the mass of the ions, and s the shear wave velocity.

When the skin depth δ is much smaller than the acoustic wavelength $\lambda = 2\pi s/\omega$, the acoustic amplitude for $z > \delta$ is given by⁹

$$|u| = H(0)H_0/4\pi ds(1 + \beta^2)^{1/2}, \quad (3)$$

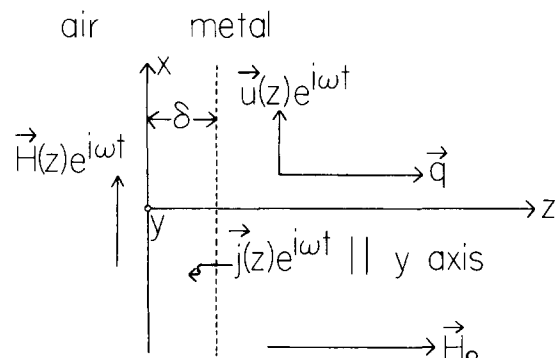


FIG. 1. Geometry for the electromagnetic generation of acoustic waves. The external rf magnetic field $\mathbf{H}(z)$ penetrates the metal to within its skin depth δ and induces eddy currents $\mathbf{j}(z)$ in that region. Since $\mathbf{H}_0$ is along the z axis, $\mathbf{j}(z)$ experiences a Lorentz force in the x direction and transverse acoustic waves of wavevector $\mathbf{q}$ are thereby generated.

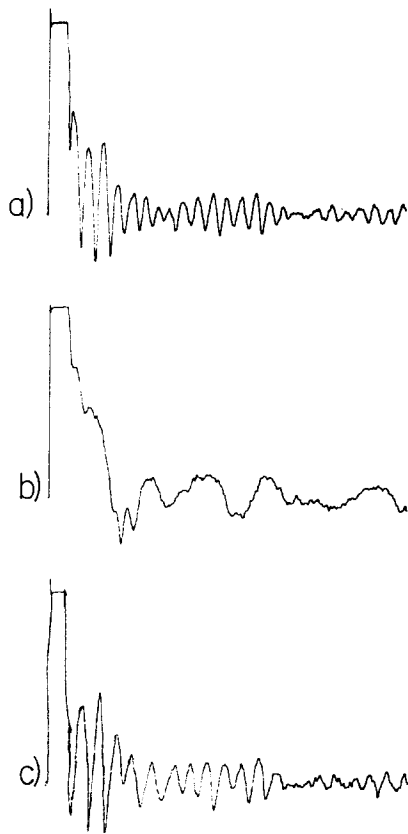


FIG. 2. The phase detected response of the NMR receiver after rf pulses near an acoustic resonance frequency of a 1.0-mm thick Al plate. The acoustic resonance is at 7.656 MHz ($m = 2Df_m/s = 5$). (a) $f = 7.600$ MHz; (b) $f = 7.640$ MHz; (c) $f = 7.690$ MHz. In each case the magnetic field H_0 is 5.6 kG and the trace length is 512 μ s.

where $H(0)$ is the rf field at the surface, $\beta = q^2\delta^2/2$, and $q = \omega/s$.

The inverse process also occurs. When the acoustic wave reaches a surface in the presence of H_0 , it carries with it an rf electric field.⁸ Since Maxwell's equations demand continuity of the tangential component of an electric field when passing from one medium to another, an electromagnetic wave is emitted from the metal. A nearby rf coil with the correct polarization will then pick up a voltage given by^{7,9}

$$V_R \propto H_0^2/ds(1 + \beta^2). \quad (4)$$

Therefore an rf coil near a metal slab in a static magnetic field can be used both to generate ultrasonic waves in the slab and to detect the echoes as they return from the opposite surface.

II. EFFECTS IN THE PROBE WALLS

It is evident that the geometry for the electromagnetic generation and detection of acoustic waves is present in pulsed NMR probes. An rf pulse across the coil will generate transverse (shear) acoustic waves in the metal probe walls lying parallel to the pole faces of the magnet. The acoustic pulses are reflected back and forth inside each wall, emitting electromagnetic waves each time a surface is reached. Thus, whenever they return to the

surface at which they were generated, a voltage will be induced across the coil.

The electromagnetic-acoustic coupling is usually weak, but the thickness D of the walls is generally on the order of the ultrasonic wavelength in metal, in which case standing acoustic waves may be set up.⁸ Since the driving force is acting only at one surface of each wall, these standing waves will occur whenever D is equal to an integer number of half wavelengths. One would then expect the response V_R of the coil to be negligible except at radio frequencies near the acoustic resonance frequencies given by

$$f_m = ms/2D, \quad (5)$$

where $\lambda f = s$ and $m = 1, 2, 3, \dots$

When the probe walls were made from 1.0-mm-thick plates of commercial Al, acoustic ringing signals were observed near six regularly spaced frequencies below 10 MHz. Figure 2 shows how the NMR receiver response changes as the radio frequency is brought through the signal corresponding to the $m = 5$ resonance. The average value of $\Delta f = f_{m+1} - f_m$ was 1.533 MHz, yielding a shear wave velocity $s = 2D\Delta f$ of 3.1×10^6 mm/s. For a 2.0-mm Al wall 13 resonances were observed, with an average Δf of 0.774 MHz and a shear wave velocity which was once again 3.1×10^6 mm/s. This value of s is consistent with other reported values for Al (3080 m/s,¹⁰ 3040 m/s¹¹).

An rf pulse of frequency f_0 and finite duration t_w has nonzero Fourier components at frequencies near f_0 . The intensity of these components is given by

$$I(f) \propto |\sin[\pi t_w(f_0 - f)]/(f_0 - f)|^2. \quad (6)$$

$I(f)$ has a central maximum at $f = f_0$, where $I(f_0) \propto t_w^2$. But it also has subsidiary maxima at $f = f_0 \pm n/2t_w$ ($n = 3, 5, 7, \dots$) with amplitudes proportional to $(f_0 - f)^{-2}$, and minima (zeros) at $f = f_0 \pm n'/t_w$ ($n' = 1, 2, 3, \dots$). Thus for a given pulse length, as one varies the radio frequency near an acoustic resonance frequency f_m the largest response will be observed when

TABLE I. The positions (in kHz) of the first five subsidiary maxima and minima on both sides of the $m = 4$ ($f_m = 6.110$ MHz) central acoustic response of the 1.0-mm Al wall. The pulse length was 33 μ s. E and C refer to the experimental and calculated values respectively. $\Delta f \equiv f_m - f_0$.

j	$\Delta f_{\min} E$	$\Delta f_{\min} C$	$\Delta f_{\max} E$	$\Delta f_{\max} C$
1	32 -32	30 -30	43 -45	45 -45
2	62 -62	61 -61	78 -78	76 -76
3	92 -93	91 -91	109 -107	106 -106
4	122 -124	121 -121	138 -138	136 -136
5	152 -154	152 -152	167 -167	167 -167

$f_0 = f_m$, and a series of successively smaller maxima in the response will occur at $f_0 = f_m \pm n/2t_w$. Table I lists the observed and calculated values of $(f_m - f_0)$ for the j th minimum and maximum acoustic responses of the 1.0-mm Al wall to radio frequencies near the $m = 4$ acoustic resonance. The pulse length was $33 \mu\text{s}$ and the j th minima and maxima were calculated from $(f_m - f_0)_{\text{min}} (\text{kHz}) = \pm j/t_w$ and $(f_m - f_0)_{\text{max}} (\text{kHz}) = \pm (2j + 1)/2t_w$, where t_w is in ms. When $(f_m - f_0)$ is held constant and nonzero, the amplitude of $I(f_m)$, and therefore the acoustic response, will oscillate as t_w is increased. Table II lists the observed and calculated values of t_w for the k th minimum $\{\sin[\pi t_w(f_m - f_0)] = 0\}$ and maximum $\{\sin[\pi t_w(f_m - f_0)] = \pm 1\}$ acoustic responses of the 1.0-mm Al wall when $m = 4$ and $f_m - f_0 = 78 \text{ kHz}$. Since the amplitude of the central response is proportional to t_w^2 rather than oscillatory in t_w , one can easily distinguish the subsidiary maxima of a given acoustic resonance response from the central maximum merely by varying the pulse length and observing the resulting change in the response.

When the pulse length is shorter than the time required for the acoustic wave to make a round trip, one observes a series of distinct echoes of width equal to that of the rf pulse. Figure 3 shows the response of the NMR receiver to acoustic echoes resulting when one wall of the probe was replaced by a 12.7-mm-thick plate of commercial Al. The signals were amplified 10 to 20 times more than those in Fig. 2. The acoustic velocity in the plate is given by $s = 2D/\tau$, where τ is the time interval between echoes. The average τ [found from Fig. 3(a), which had the sharpest echoes] was $8.29 \mu\text{s}$, giving a shear wave velocity of $3.06 \times 10^6 \text{ mm/s}$.

These investigations clearly indicate that the spurious ringing signals due to the metal surfaces of the probe are a result of acoustic standing wave resonances being set up inside the walls. The exact dependence of the amplitude of these signals on H_0 , H_1 , and t_w was not determined; however, the qualitative features are consistent with this explanation. Equations (3) and (4) predict that V_R should increase as H_0 and H_1 ¹² are increased. The amplitude at resonance should also increase with the length of time the driving force is applied, at least within the range of pulse lengths studied here (less than $100 \mu\text{s}$). Since $\beta \propto f^2$ one might expect the amplitude of the re-

TABLE II. The pulse lengths below $100 \mu\text{s}$ (t_w in μs) at which subsidiary maxima and minima occur in the $m = 4$ acoustic response of the 1.0-mm Al wall. $f_m = 6.110 \text{ MHz}$ and $f_0 = 6.032 \text{ MHz}$. E and C refer to the observed and calculated values, respectively.

k	$t_{w,\text{max}}' E$	$t_{w,\text{max}}' C$	$t_{w,\text{min}}' E$	$t_{w,\text{min}}' C$
1	8	6	14	13
2	21	19	27	26
3	33	32	39	38
4	46	45	52	51
5	59	58	64	64
6	71	71	77	77
7	84	83	90	90
8	96	96	—	—

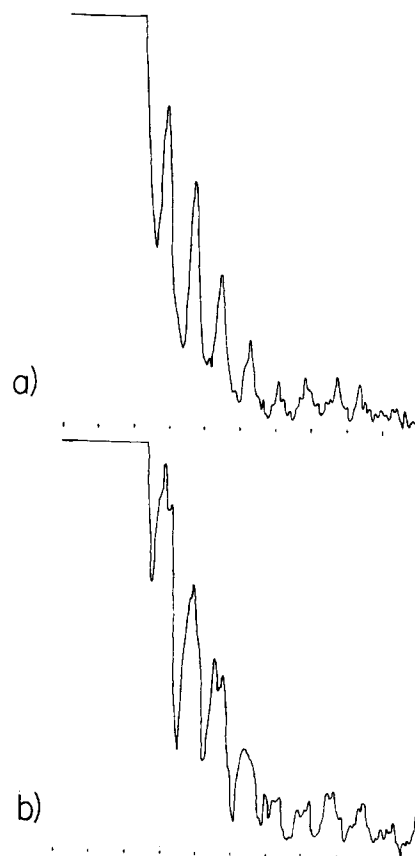


FIG. 3. Acoustic echoes exhibited when one probe wall is replaced by a 12.7-mm slab of Al. The other wall is brass in order to eliminate its contribution to the signal. The receiver response was not phase detected. In both cases $H_0 = 10 \text{ kG}$, $f = 6.797 \text{ MHz}$ and the horizontal scale is $10 \mu\text{s}$ per div, but the rf pulse length in (a) is $3 \mu\text{s}$ and in (b) $5 \mu\text{s}$.

sponse to have, in addition to the frequency dependence of $I(f)$ given by Eq. (6), a further dependence on frequency due to the factor $(1 + \beta^2)^{-1}$ in Eq. (4). However, for Al at room temperature, $\beta^2 = 0.014$ at 10 MHz ,⁹ so that $(1 + \beta^2)^{-1}$ is essentially constant, particularly over the frequency region of any single resonance. The response amplitude is smaller for the higher frequency resonances [larger values of m in Eq. (5)], but that is because shorter wavelengths are more sensitive to variations in the thickness of the plate.

This explanation suggests four lines of approach for alleviating the problem:

(1) Frequency and geometry manipulations. Since the amplitude of the acoustic ringing is large only when operating at or near one of the acoustic resonance frequencies of the probe walls, the problem can often be avoided merely by operating as far as possible away from these frequency regions. The resonance frequencies can be determined by actually searching for them or, when s and D are known, by using $f_m = ms/2D$. Conversely, if one has little flexibility in choosing the spectrometer frequency, the wall thickness can be chosen such that there are no acoustic resonances in the frequency range to be used. This technique begins to lose its effectiveness when $1/t_w$ is on the order of Δf . In that case the pulse will contain

frequency components near one or more of the resonances regardless of the spectrometer frequency used.

(2) Shielding the probe walls from the rf field. Speight *et al.*² reported success in eliminating the acoustic ringing by inserting a grounded copper wire shield between the coil and the conducting surface of the probe. A similar technique, in which the coil was surrounded by a copper screen which was grounded to the probe box, was effective in our investigations. The resulting suppression of the ringing is shown in a comparison of Figs. 4(a) and 4(b).

(3) Damping the acoustic waves. This can be accomplished by coating the metal surfaces with an acoustically lossy substance. When the silicone rubber compound RTV was used, both the duration and amplitude of the ringing were greatly reduced, as shown in Fig. 4(c). An RTV coating on the silvered wall of a Dewar was also effective in suppressing the acoustic responses found there at room temperature.

(4) Choice of probe material. Two factors enter into how effective a particular metal will be at producing the acoustic ringing effect. First is its conversion efficiency for the electromagnetic generation of acoustic waves, and second is its efficiency for transmitting the acoustic waves. The former was calculated as a function of frequency for several metals at room temperature by Dobbs and Llewellyn.¹³ They assumed constant H_0 and $H(0)$ and negligible acoustic attenuation within the metal. Their calculations indicate that the relative efficiency for the electromagnetic generation of shear waves in brass as compared to Al in the frequency range 2–18 MHz varies from approximately –7.8––9.7 dB. Brass is also a much poorer transmitter of ultrasonic waves due to scattering by metal grains.¹⁴ Both factors indicate that probe walls made of brass should exhibit less acoustic ringing than those made of Al. Figure 4(d) shows the receiver response under conditions identical to those in

2(a), except that the Al walls have been replaced with brass, and the signal gain and time scale are greater by factors of four and two, respectively. Since the shear wave velocity is different in brass than in Al, one would generally not expect both types of walls to have resonances at the same frequencies. The acoustic resonance frequencies for the brass plates were calculated using the plate thickness and reported shear wave velocities, but no detectable ringing was observed in the frequency regions around these values. Therefore, at least at the magnetic field strengths used in our investigations (<10 kG), the acoustic ringing problem in brass walls is negligible.

Equation (4) predicts that $V_R \propto H_0^2$, so at fields much higher than 10 kG it is possible that the ringing may show up even in brass. At very high fields, combinations of all four techniques might be required to suppress the ringing sufficiently. On the other hand, the problem should be reduced somewhat by going to higher frequencies. Three factors are involved in this case: (1) as mentioned previously, the amplitude of the resonance signal is smaller for larger m , (2) the conversion efficiency is reduced at higher frequencies,^{9,13} and (3) particularly in the case of brass, the shorter wavelengths will be scattered more.¹⁵

III. COIL EFFECTS

Even when the acoustic ringing within the probe walls has been eliminated, one can often observe similar signals at a number of different frequencies. In several cases, the probe walls were removed entirely and significantly large ringing effects were observed at several frequencies below 10 MHz. The amplitudes of these residual signals are once again dependent upon H_0 , H_1 , and t_w . The large thickness of the magnet pole faces makes it unlikely that ultrasonic resonances could be set up there. We have therefore attributed this residual ringing to acoustic responses in the coil itself, as originally suggested by Clark.¹

It is reasonable to suppose that a similar effect can happen in the wire of the coil as occurs in the probe walls. In this case the Lorentz force acts directly on the rf current flowing at the surface of the wire. This force will be approximately parallel to the coil axis at all points along the wire. Under these conditions one can easily imagine both transverse and longitudinal acoustic waves propagating across any given section of the wire, as shown in Fig. 5. Since the wave velocities of the two modes are not the same, each would have a different set of resonance frequencies. The situation is further complicated by the possibility of still other modes of propagation, such as surface waves and wave guide modes. Unfortunately, each coil is tunable over a limited frequency range, so a complete acoustic resonance spectrum for any given coil cannot be obtained merely by sweeping the radio frequency. In view of these complications, no serious attempt was made to determine which acoustic modes were responsible for the residual signals.

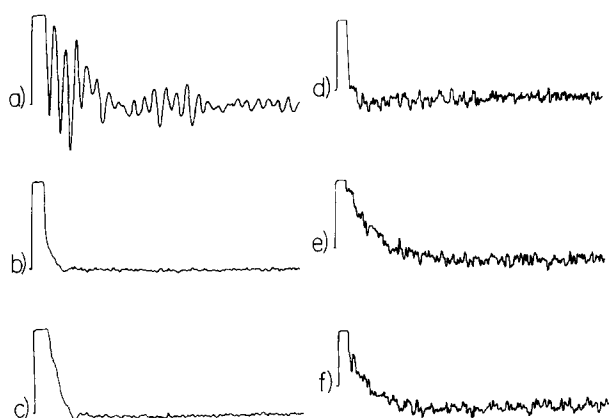


FIG. 4. (a) The phase detected response of the NMR receiver after an rf pulse at 7.700 MHz in an untreated Al probe. $H_0 = 5.6$ kG and the trace length is 512 μ s; (b) same as (a), but with copper screen shield in place; (c) same as (a), but the walls are RTV coated (d) response when probe walls are brass. The signal has been amplified a factor of four greater than those in (a) through (c) and the trace length is now 1024 μ s; (e) same as (d), but showing the FID signal from ^{11}B in KBF_4 ; (f) the same FID signal when the probe walls are RTV-coated Al. All traces were taken at room temperature.

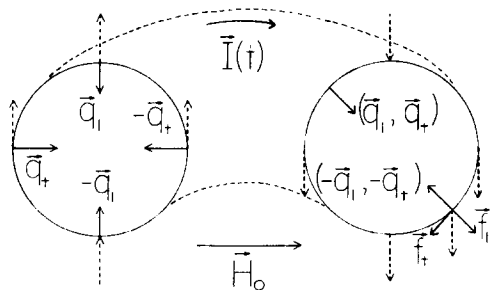


FIG. 5. Geometry for the generation of transverse (q_t) and longitudinal (q_l) acoustic waves inside the coil wire. The view is of a cross-section of one loop of the coil. Dashed vectors represent the Lorentz force $\mathbf{F}(t) = \mathbf{I}(t) \times \mathbf{H}_0$ acting at the surface of the wire. Pure transverse waves will propagate parallel to $\mathbf{H}_0$ [the Lorentz force $\mathbf{F}(t)$ is tangential to the surface] and pure longitudinal waves perpendicular to $\mathbf{H}_0$ [$\mathbf{F}(t)$ normal to surface]. For intermediate orientations, $\mathbf{F}(t)$ has both tangential (f_t) and normal (f_l) components, so that mixtures of the two modes will propagate.

It was observed that coils wound from finely stranded insulated wires did not exhibit the type of ringing characteristic of solid-wire coils. In most cases, few if any residual signals were observed in the stranded wire, and those that did show up were at least an order of magnitude smaller than those found in solid wire. Also, there was no clear resonance structure to the signals, but rather the amplitudes were fairly constant over a broad frequency range (at least 1 or 2 MHz). This can be explained by the fact that the diameter of each strand is smaller than the ultrasonic wavelengths below 10 MHz, so that normal acoustic resonances cannot occur. Indeed, Clark¹ originally suggested the use of stranded wire as a possible solution to the coil disease.

The acoustic responses of the coil can be more difficult to eliminate than those of the probe walls. Coating the wire with RTV appeared to have very little effect on the ringing, but there might exist other acoustically lossy substances which would be more successful. Besides operating at frequencies where no such effects occur, the best solutions seem to be the use of finely stranded insulated wires or solid wire of a different diameter than that from which the problem coil was made. By any or all of these methods, the ringing can generally be reduced to a negligible level.

Finally, if all attempts to eliminate the acoustic ringing in either the probe walls or coil are unsuccessful, the possibility still remains of using computer techniques for subtracting out the unwanted signals. For example, when signal averaging the FIDs from a large number of pulses one might simply run the signal averager in the "add" mode with the sample in place and then run in the "subtract" mode, for an equal number of pulses, with the sample removed. The disadvantage of this technique is that it of course increases the data acquisition time by a factor of two.

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